Soong Shin Ee

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Education

University of Southampton| United Kingdom *September 2021 - June 2023*
University of Southampton| Malaysia *September 2019 – August 2021*
MEng Electrical and Electronic Engineering (Hons) Second Class Honours (Upper Division)
Relevant Modules: Digital Systems and Signal Processing, Applied Control Systems, Control System Design, Nonlinear Control of Aerospace Systems, Numerical Methods, Integral Transform Methods, Biologically Inspired Robotics

Skills

Programming: C, C++, Embedded C, Python, Verilog
Firmware & Embedded Systems: Arduino, ModelSim, Quartus, LabView, myRIO
Software: EAGLE, MATLAB, LTspice, Siemens PCS7, Siemens TIA Portal, Ignition, AutoCAD
Hardware Proficiency: PLC/SCADA Systems, Siemens S7-1500/ S7-400 series
Languages: English, Chinese, Malay, Cantonese

Professional Experience

System and Project Engineer | Pro Cahaya Industries Sdn Bhd *November 2023 - Present*

Infineon Technologies Kulim | *November 2023 - Present*
Project Involved: Gas Monitoring System (GMS), Continuous Quality Control System (CQC)
Scope of work:

- Designed and implemented control systems, including PLC/SCADA and system architecture development
- Coordinated project planning and client meetings, ensuring timely delivery of automation system integration solutions
- Preparation and submission of technical documentation, including Panel Drawings and System Architecture Diagrams
- Conducted Factory Acceptance Tests (FAT) and Site Acceptance Test (SAT) at project sites

Johor River Water Works (JRWW) – PUB | *November 2024 – December 2024*
Scope of work:

- Performed loop test, commissioning and troubleshooting of electrical and control systems on-site
- Interpreted and implemented electrical drawings for PLC panels and instrument connections

Engineering Experience

Implementation of Cooperative Control Algorithms in Collaborative Robots | Individual Project

- Researched and derived mathematical models for consensus and flocking control algorithms
- Implemented and simulated control algorithms using MATLAB and analyzed system performance using MATLAB Simulink

Digital Systems and Signal Processing

- Designed and implemented RTL synthesis for combinational and sequential systems using System Verilog
- Designed and synthesized a MIPS microprocessor on FPGA hardware, including hardware-software design and verification
- Analyzed and designed Finite State Machine (FSM) and Synchronous State Machines (SSM) for control logic applications
- Applied signal processing techniques (Fourier Transform, Digital Filters) using MATLAB to analyze system performance

Advanced Control System Design and Implementation for Robotics and Servo Systems

LEGO Mindstorms NXT Robot | Software Design and Simulation

- Designed and optimized Proportional-Integral-Derivative (PID) and Model Predictive Control (MPC) controller in MATLAB
- Analyzed and improved above controller performance for robot system

Quanser QUBE-Servo System | Hardware Implementation

- Implemented control system designs on hardware using LabView and myRIO software
- Designed and fine-tuned Linear Quadratic Regulator (LQR), Model Predictive Control (MPC) and Iterative Learning Control (ILC) Systems for optimized hardware performance

Formation and Swarm Behavior of Remotely Operated Vehicles (ROV)

- Implemented a Python-based framework to evaluate robotic swarm behavior and mission cohesion
- Analyzed system performance using graphical tools (e.g. heat maps, dual-axis graphs)